DA3D: Domain-Aware Dynamic Adaptation for All-Weather Multimodal 3D Detection

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Abstract

LiDAR-Radar fusion has been widely regarded as an effective strategy for enhancing sensor-level robustness in 3D perception under adverse weather. However, it remains fundamentally insufficient to address feature-level domain shifts induced by diverse weather conditions — a critical yet often overlooked bottleneck in multimodal 3D object detection. In this work, we advocate a new perspective: all-weather 3D detection should be formulated as a lightweight capacity allocation problem, rather than simply enlarging or duplicating models for each weather domain. To this end, we propose DA3D, a Domain-Aware Dynamic Adaptation framework that leverages LoRA as a domain-adaptive capacity controller for efficient and scalable feature modulation. In addition, we introduce a domain-aware rank adaptation strategy that dynamically reallocates LoRA capacity based on domain difficulty, allowing the model to focus its representational power where it matters most. Extensive experiments on the K-Radar benchmark show that DA3D consistently improves 3D detection across both radar-only and LiDAR-Radar fusion backbones, achieving +4.9% AP_{3D} on RTNH, +3.8% on 3D-LRF, and +8.1% on L4DR at IoU=0.5. Notably, DA3D outperforms existing multi-weather modeling methods under the same parameter budget, offering a practical and scalable solution for robust all-weather 3D perception. The code is available at https://github.com/Dawns14/DA3D.

CCS Concepts

• **Computing methodologies → Object detection**.

Keywords

3D Object Detection, Multimodal Fusion, Low-Rank Adaptive, Multi-Domain

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1 Introduction

Accurate and robust 3D object detection is fundamental to autonomous driving, where perception systems must operate reliably across diverse and adverse environments. LiDAR and radar sensors have emerged as the two most widely used modalities in 3D perception, each with distinct strengths and limitations. LiDAR provides high-resolution geometric information, which is critical for precise object localization in favorable weather [4, 6, 13, 39]. However, its performance may degrade in adverse weather conditions such as rain, fog, or snow due to increased noise and reduced sensing range. In contrast, 4D radar offers reliable long-range perception which is less affected by weather variations [1, 9, 10], enabling radar-only approaches to be more attractive for real-time or cost-sensitive applications [16, 25, 31], although this comes at the cost of limited angular resolution and reduced localization accuracy. To overcome these limitations, recent studies have focused on LiDAR-Radar fusion frameworks which combine the complementary strengths of

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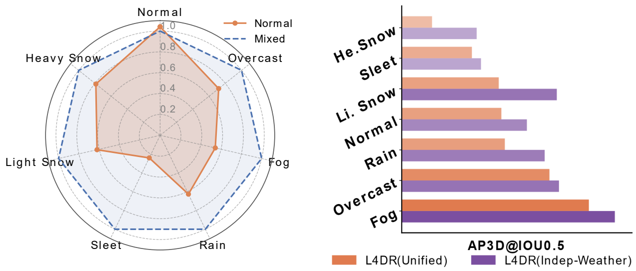

(a) Domain Shift Across Weather Conditions **(b) Unified vs. Indep-Weather Model**

Figure 1: (a) Relative $\mathrm{AP}_{\mathrm{3D}}$ of the LiDAR–4D Radar fusion model L4DR [14], comparing the *Normal* setting (trained on normal weather only) against the *Mixed* setting (trained on all weather conditions). Values are computed as the performance drop of Normal relative to Mixed across seven weather types. (b) Accuracy–parameter trade-off between unified and independent-weather models: while weather-specific models improve detection, they incur substantial parameter overhead.

these two modalities. In LiDAR-Radar fusion, LiDAR provides accurate geometric details for precise localization, while 4D radar enhances robustness under adverse weather and long-range scenarios. Fusion-based methods such as InterFusion [28], 3D-LRF [7], and L4DR [14] have demonstrated superior detection performance, particularly in complex urban environments involving occlusions, multiple objects, and challenging weather conditions.

While 4D radar and LiDAR-Radar fusion enhance sensor resilience under adverse weather, contemporary 3D detectors remain fundamentally limited by their weather-agnostic architectural designs. Existing approaches, no matter radar-only [16, 25, 31] or fusion-based [7, 14, 28], typically deploy a single unified model which is trained on aggregated multi-weather data, implicitly assuming that shared representations suffice for cross-domain generalization. However, this assumption is challenged by growing empirical evidence. Zhang et al. [36] report notable feature distribution shifts across weather conditions, despite the perceived robustness of 4D radar. As shown in Fig.1(a), our analysis further corroborates this finding by showing that even the state-of-the-art L4DR fusion model [14] suffers significant performance degradation in unseen weather conditions. A straightforward solution is to train specialized models for each weather type (Fig. 1(b)), which can effectively mitigate domain shifts and improve detection accuracy under diverse weather conditions. However, it leads to poor scalability: supporting seven conditions requires approximately 7× model parameters (~483M total), leading to prohibitive computational and memory overhead. This exposes a critical dilemma: **how can we design parameter-efficient models that dynamically adapt to weather-induced domain shifts while preserving unified inference capability and scalability?**

To address these issues, we introduce **DA3D**, a Domain-Aware Dynamic Adaptation framework for all-weather 3D object detection. The core motivation behind DA3D is the observation that weather-induced domain shifts in sensor data are often structured and low-dimensional, such as long-range point attenuation in fog or dense local noise in snow. This motivates us to leverage LoRA [12], not as a generic fine-tuning method, but as a principled mechanism to model domain-specific feature variations with minimal parameter overhead. Crucially, unlike conventional usage in language models, we reinterpret LoRA in 3D perception as a **domain-adaptive capacity controller**, allocating flexible representational power across weather domains while preserving a unified backbone. By injecting lightweight, weather-aware adapters into existing 3D detection architectures, DA3D achieves efficient and scalable adaptation to diverse weather conditions without modifying the model structure or duplicating parameters.

Beyond domain-specific modeling, a key contribution of DA3D is the introduction of a domain-aware rank adaptation module that dynamically reallocates representational capacity across different weather domains. While LoRA facilitates efficient domain-specific feature adaptation, assigning equal rank capacity to all domains proves suboptimal due to the varying severity of domain shifts under different weather conditions. To address this issue, we design an adaptive rank allocation strategy to balance model capacity across weather domains. It removes redundant ranks from easier domains and reallocates them to harder ones, guided by gradient sensitivity and SVD-based reconstruction errors. This mechanism enables DA3D to concentrate model capacity where domain shifts are more severe, thereby achieving robust performance under a fixed-parameter budget.

Extensive experiments on the K-Radar benchmark [25] demonstrate the superior effectiveness and generality of DA3D on the radar-only and LiDAR-Radar fusion backbones. DA3D consistently improves AP_{3D} at IoU=0.5 by 4.9% on RTNH[25], 3.8% on 3D-LRF [7], and 8.1% on L4DR [14], while introducing only around 30% additional parameters over the base models. These results highlight that DA3D serves as a plug-and-play, architecture-agnostic solution for robust and scalable 3D object detection in diverse and adverse weather conditions.

The main contributions of this paper are summarized as follows:

- We identify and address the overlooked domain shift problem in all-weather 3D object detection, which is often underestimated by existing methods assuming the robustness of LiDAR-Radar fusion models under adverse weather conditions.
- We propose **DA3D**, a Domain-Aware Dynamic Adaptation framework. To the best of our knowledge, we are the first to introduce LoRA as a domain-adaptive capacity controller for 3D perception, enabling efficient modeling of weather-specific variations without modifying the backbone architecture.
- We introduce a domain-aware rank adaptation module that addresses the domain difficulty imbalance across weather conditions by dynamically reallocating LoRA capacity, enabling robust performance under a fixed-parameter budget.
- Extensive experiments on the K-Radar benchmark show that DA3D consistently outperforms state-of-the-art multi-weather modeling methods across both radar-only and LiDAR–Radar fusion backbones. DA3D achieves the best performance with comparable parameter budgets.

2 Related Work

2.1 Multimodal Detection in Adverse Weather

3D Object Detection is a fundamental task in autonomous driving systems, which aims to acquire accurate locations of nearby objects in 3D representation space [3, 24]. Adverse weather conditions, such as fog, rain, and snow, always degrade the detector's performance, which seriously hinders the safety and reliability of autonomous driving systems [15, 35]. Among object detection algorithms in adverse weather conditions, image-based methods [20, 23, 30] often suffer from significant performance degradation due to reduced visibility and image quality, and LiDAR-based methods [5, 18, 19, 26] deteriorate due to signal noise, reduced sensing range, and sparse point clouds. To overcome the individual limitations of each sensor, multimodal detection, such as radar LiDAR fusion, has attracted increasing attention in recent years. Early fusion models [21, 29] primarily focused on using radar to complement LiDAR by providing velocity and range information. More recent frameworks have integrated 4D radar, which captures both spatial and Doppler velocity information, further improving detection in dynamic, low-visibility conditions. Notable approaches such as InterFusion [28], 3D-LRF [7], and L4DR [14] propose unified fusion architectures to jointly process LiDAR and radar features, demonstrating substantial performance gains in challenging environments. However, despite recent progress, most fusion models remain weather-agnostic, relying on a unified architecture trained over mixed-weather datasets. This naive training strategy implicitly assumes that shared representations suffice for cross-weather generalization, which conflicts with growing evidence of severe domain shifts caused by varying weather conditions [36], highlighting the urgent need for domain-aware modeling in all-weather 3D detection.

2.2 LoRA for Multi-Task and Multi-Domain

Low-Rank Adaptation (LoRA)[12] and its variants[17, 22, 37, 38] have emerged as a powerful family of parameter-efficient fine-tuning techniques, enabling effective adaptation of large language models (LLMs) and vision models with minimal trainable parameters. By freezing the pretrained backbone and injecting learnable low-rank matrices into selected layers, LoRA significantly reduces trainable parameters while preserving performance. Recent research has extended LoRA to multi-task learning, enabling a single backbone to support multiple tasks with compact task-specific modules. MTL-LoRA [33] improves multi-task LoRA by introducing a Mixture-of-Experts design with task-specific routing for flexible parameter sharing, while MTLoRA [2] further optimizes capacity allocation and reduces task interference. Beyond multi-task learning, LoRA has also been applied to multi-domain adaptation across various applications, including domain transfer in-depth estimation [32], recommendation systems [27], and CTR prediction [34], where domain differences are primarily user-driven or task-driven.

In contrast to these scenarios, we address a fundamentally different and practically challenging problem setting: domain adaptation for weather-induced feature shifts in multimodal 3D object detection. Unlike prior LoRA-based methods that focus on user-driven or task-driven domains, we target sensor-level domain shifts arising from environmental factors such as rain, fog, and snow, which directly affect the feature distributions of LiDAR and radar signals. Moreover, we propose a domain-aware rank adaptation module to dynamically allocate LoRA capacity based on domain difficulty. To the best of our knowledge, this is the first work that leverages LoRA as a domain-adaptive capacity controller in 3D perception, tackling the critical yet underexplored challenge of all-weather multimodal 3D object detection.

3 Methodology

In this section, we first describe the multimodal 3D object detection backbone and review the standard LoRA approach (Section 3.1). Then we introduce our proposed DA3D architecture with domain-specific LoRA branches in Section 3.2. Finally, we present the domain-aware dynamic rank adaptation mechanism that reallocates LoRA capacity across weather conditions (Section 3.3).

3.1 Preliminaries

Multimodal LiDAR-Radar Backbone. Our baseline is a LiDAR–Radar detection backbone that supports both radar-only and fused LiDAR–Radar inputs. As shown in Fig. 2(a), it follows a two-stage feature extraction: an initial 3D convolutional backbone $\mathcal{F}_{3D}$ processes the sparse point cloud (voxelizing points from LiDAR and radar into a 3D grid), and a 2D convolutional backbone $\mathcal{F}_{2D}$ further processes the bird's-eye-view (BEV) feature map produced by the 3D encoder. Let P denote the set of LiDAR points (if available) and R the radar returns; The overall backbone $f(P, R; W)$, parameterized by $W = \{\mathcal{F}_{3D}, \mathcal{F}_{2D}\}$, yields a deep scene representation $F = f(P, R; W)$. This representation is then fed to a detection head $g(\cdot)$ that outputs 3D bounding boxes. In a radar-only setup, P is empty and the same backbone $f(\cdot)$ uses radar features to produce F, ensuring a unified architecture for both sensor configurations.

By default, both $\mathcal{F}_{3D}$ and $\mathcal{F}_{2D}$ are trained jointly on data from all weather conditions, following the common practice of mixed-domain training. In the following sections, we insert adaptation modules into both $\mathcal{F}_{3D}$ and $\mathcal{F}_{2D}$ to enable domain-specific feature modulation.

LoRA. LoRA [12] is a widely adopted parameter-efficient fine-tuning technique based on the low-rank decomposition of weight updates. Given a pretrained weight matrix $W^{(l)} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ at layer l (either fully connected or convolutional), the goal of fine-tuning is to learn an update $\Delta W^{(l)} = W'^{(l)} - W^{(l)}$ that adapts the model to a downstream task. Prior studies have shown that such updates often lie in a low-dimensional subspace, and thus $\Delta W^{(l)}$ can be approximated by the product of two low-rank matrices:

$$\Delta W^{(l)} = B^{(l)} A^{(l)}, \quad B^{(l)} \in \mathbb{R}^{d_{\text{out}} \times r}, \quad A^{(l)} \in \mathbb{R}^{r \times d_{\text{in}}} \tag{1}$$

To ensure zero-initialization of $\Delta W^{(l)}$ at the start of training, $A^{(l)}$ is initialized with a Gaussian distribution and $B^{(l)}$ with zeros. During training, the original weight $W^{(l)}$ remains frozen, and only $A^{(l)}$ and $B^{(l)}$ are optimized. For an input feature vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$, the output $\mathbf{h} \in \mathbb{R}^{d_{\text{out}}}$ is computed as:

$$\mathbf{h} = W^{(l)}\mathbf{x} + B^{(l)}(A^{(l)}\mathbf{x}) \tag{2}$$

which introduces negligible additional cost while enabling effective adaptation through low-rank projections.

While LoRA has been widely adopted for parameter-efficient fine-tuning, its potential for domain shift adaptation in multi-domain 3D

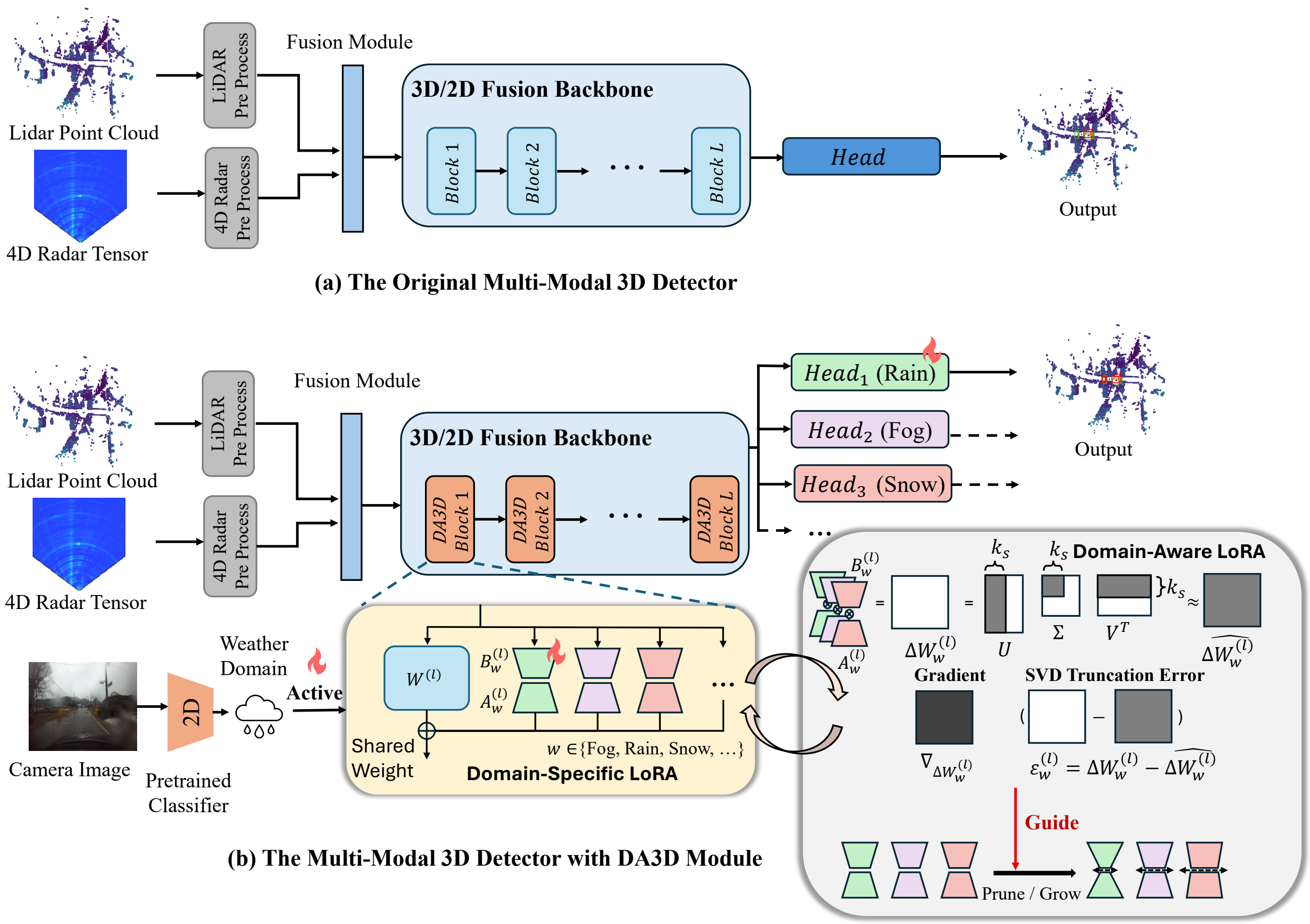

Figure 2: Overview of DA3D. Given multimodal inputs from LiDAR and 4D Radar, domain-specific LoRA modules are inserted into both the 3D sparse encoder and 2D BEV encoder for domain-aware feature modulation. At runtime, a weather classifier selects the active domain branch, while a dynamic rank adaptation strategy adjusts the LoRA capacity per domain and per layer.

perception remains underexplored. We posit that weather-induced domain shifts often exhibit structured, low-dimensional patterns (e.g., point sparsity in fog, noisy reflections in snow), making them particularly well-suited for LoRA-based modeling with minimal parameter overhead. This motivates assigning a lightweight, domain-specific LoRA module per weather condition, enabling efficient and modular adaptation without altering the shared backbone, and incurring far lower overhead than training separate models. More importantly, this design enables scalable Multimodal 3D detection under diverse weather conditions.

3.2 Domain-Specific LoRA

As illustrated in Fig. 2(b), DA3D inserts domain-specific LoRA modules into all convolutional layers of both the 3D sparse encoder and the 2D BEV encoder, enabling domain-aware feature modulation under different weather conditions. During inference, a weather classifier dynamically selects the active domain branch to adapt to the current environment.

Problem Formulation. Given the multimodal 3D detection backbone $f(\cdot)$ introduced in Section 3.1, our goal is to enable domain-aware adaptation to diverse weather conditions while maintaining a unified model structure. Let $\mathcal{W} = \{\texttt{normal}, \texttt{rain}, \texttt{fog}, \texttt{snow}, \dots\}$ denote the set of predefined weather domains. For each domain $w \in \mathcal{W}$, we aim to learn domain-specific low-rank residuals to modulate the shared backbone weights, enabling domain-conditioned feature adaptation without duplicating the entire model.

Domain-Specific LoRA Modules. To achieve this, we insert a set of domain-specific LoRA modules into all convolutional layers of both the 3D sparse encoder $\mathcal{F}_{3D}$ and the 2D BEV encoder $\mathcal{F}_{2D}$. For each convolutional kernel tensor $W^{(l)} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k_1 \times \cdots \times k_n}$ at the l-th layer, we apply a domain-specific low-rank residual in the flattened space of $W^{(l)}$ without modifying its original structure. Specifically, we reshape a learnable low-rank residual $\Delta W_w^{(l)}$

into the same shape as $W^{(l)}$ and add it to the original weight for domain adaptation. The domain-adapted weight $W_w^{(l)}$ constructed as follows:

$$\Delta W_w^{(l)} = \text{reshape}(B_w^{(l)} A_w^{(l)}) \tag{3}$$

$$W_w^{(l)} = W^{(l)} + \Delta W_w^{(l)} \tag{4}$$

where $A_w^{(l)} \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B_w^{(l)} \in \mathbb{R}^{d_{\text{out}} \times r}$ operate in a flattened space with $d_{\text{in}} = C_{\text{in}} \times k_1$ and $d_{\text{out}} = C_{\text{out}} \times k_2 \times \cdots \times k_n$. The original kernel $W^{(l)}$ remains in its native convolutional shape throughout training and inference. This design balances adaptation flexibility and parameter efficiency. By learning low-rank residuals in a flattened space, we minimize the domain-specific parameter overhead while preserving the original convolutional structure. This is crucial for 3D detection backbones with high-dimensional kernels, where full-rank per-domain adaptation would incur prohibitive memory and computation costs.

Domain Routing Mechanism. To dynamically adapt to changing weather conditions during driving, we employ an auxiliary weather classifier $h(\cdot)$ that predicts the current domain label $\hat{w} \in \mathcal{W}$ from the front-view camera image. The classifier uses a ResNet-18 [11] backbone pretrained on ImageNet [8] and fine-tuned on the K-Radar dataset [25], achieving 99.2% test accuracy. During inference, the predicted label $\hat{w}$ is used to select the corresponding LoRA parameters $\Delta W_{\hat{w}}^{(l)}{}_{l=1}^{L}$ for weather-aware feature modulation in the backbone.

Domain-Specific Detection Heads. In addition to feature adaptation, we adopt domain-specific detection heads g_w, where each head g_w is optimized independently for samples from domain w. At inference time, both the feature extraction modules (with $W_{\hat{w}}$) and the detection head $g_{\hat{w}}$ are activated based on the predicted domain label $\hat{w}$, enabling end-to-end domain-aware 3D object detection:

$$\hat{y} = g_{\hat{w}}(f(P, R; W_{\hat{w}})) \tag{5}$$

where (P, R) denotes the LiDAR and radar input, and $\hat{y}$ is the predicted 3D bounding boxes.

3.3 Domain-Aware Dynamic Rank Adaptation

We aim to dynamically allocate the LoRA rank $r_w^{(l)}$ for each domain-specific residual $\Delta W_w^{(l)}$ at layer l, adapting the parameter capacity to domain-specific difficulty under a total budget $R_{\text{total}}^{(l)}$, where $R_{\text{total}}^{(l)} = \sum_w r_w^{(l)}$ denotes the sum of ranks across all domains at layer l. We define k_s as the number of truncated singular values, and k_p, k_g as the number of domains to prune and grow per adaptation cycle, respectively.

Domain Importance Estimation. For each domain-specific LoRA residual $\Delta W_w^{(l)} = B_w^{(l)} A_w^{(l)}$, we first perform reduced SVD as $\Delta W_w^{(l)} = U \Sigma V^\top$, where $U \in \mathbb{R}^{d_{out} \times r_w^{(l)}}$, $V \in \mathbb{R}^{d_{in} \times r_w^{(l)}}$, and $\Sigma \in \mathbb{R}^{r_w^{(l)} \times r_w^{(l)}}$ contains singular values in descending order. We truncate the smallest k_s singular values in Σ by setting them to zero, forming Σ_{trunc}, and reconstruct the truncated weight as $\widehat{\Delta W_w^{(l)}} = U \Sigma_{\text{trunc}} V^\top$. Let $\nabla_{\Delta W_w^{(l)}}$ denote the backpropagated gradient with respect to the LoRA residual parameters at layer l for domain w. The domain importance score is then computed as:

$$e_w^{(l)} = \left| \nabla_{\Delta W_w^{(l)}} \odot \left(\Delta W_w^{(l)} - \widehat{\Delta W_w^{(l)}} \right) \right|_F^2. \tag{6}$$

Algorithm 1 Domain-Aware Dynamic Rank Adaptation

Input: Multimodal inputs: (P, R) from LiDAR and Radar, weather image I; LoRA ranks $\{r_w^{(l)}\}$; adaptation interval K; dynamic adaptation end epoch Ada_end
Output: 3D detection results $\hat{y}$
for each training epoch **do**
  Predict weather domain label $\hat{w} = h(I)$
  for each layer l in both the 3D and BEV encoder **do**
    Select domain-specific LoRA residual $\Delta W_{\hat{w}}^{(l)}$
    Modulate feature weights as $W_{\hat{w}}^{(l)} \leftarrow W^{(l)} + \Delta W_{\hat{w}}^{(l)}$
  end for
  Extract feature $F = f(P, R; \{W_{\hat{w}}\})$
  if epoch mod $K = 0$ and epoch $\leq Ada\_end$ **then**
    Perform Domain-Aware Dynamic Rank Adaptation:
      Estimate domain importance $e_w^{(l)}$ via Eq. (6)
      Prune low-importance domains via Eq. (7)
      Grow ranks for important domains via Eq. (8)
  end if
  Predict 3D bounding boxes $\hat{y} = g_{\hat{w}}(F)$
end for

Rank Pruning and Re-Factorization. We select the bottom k_p domains with the smallest $e_w^{(l)}$, and reduce their ranks as $r_w^{(l)} \leftarrow r_w^{(l)} - k_s$. We update $\Delta W_w^{(l)} = U \Sigma_{\text{trunc}} V^\top$, and re-factorize LoRA parameters as:

$$B_w^{(l)} = U \Sigma_{\text{trunc}}^{1/2}, \quad A_w^{(l)} = \Sigma_{\text{trunc}}^{1/2} V^\top. \tag{7}$$

The released budget is $R_{\text{remain}}^{(l)} = k_s \times k_p$.

Rank Growing. We select the top k_g domains with the largest gradient magnitude $g_w^{(l)} = \|\nabla_{\Delta W_w^{(l)}}\|_F^2$ to form the growing set $\mathcal{W}_{\text{grow}}$. The remaining budget $R_{\text{remain}}^{(l)}$ is allocated proportionally as:

$$r_w^{(l)} \leftarrow r_w^{(l)} + \frac{g_w^{(l)}}{\sum_{w' \in \mathcal{W}_{\text{grow}}} g_{w'}^{(l)}} \times R_{\text{remain}}^{(l)}, \quad \forall w \in \mathcal{W}_{\text{grow}}. \tag{8}$$

New rows/columns in $B_w^{(l)}$ and $A_w^{(l)}$ are initialized from a Gaussian distribution. The overall procedure of our DA3D framework is summarized in Algorithm 1. Such a domain-aware rank adaptation strategy allows DA3D to allocate LoRA capacity dynamically per domain and layer, achieving a favorable trade-off between model flexibility and parameter efficiency for all-weather 3D detection, without increasing the backbone complexity.

4 Experiments

4.1 Experimental Setup

Dataset and Evaluation. We conduct experiments on the K-Radar dataset [25], a large-scale multimodal autonomous driving benchmark with synchronized LiDAR, RGB camera, and 4D radar data

Table 1: Comparison of AP_{3D} and AP_{BEV} at IoU threshold 0.5 on the K-Radar benchmark. We report the results of baseline models and DA3D on three backbones: RTNH (radar-only), 3D-LRF (LiDAR–Radar fusion), and L4DR (LiDAR–Radar fusion) across seven weather conditions.

Method	Modality	Metric	Total	Normal	Overcast	Fog	Rain	Sleet	Li.Snow	He.Snow	Param(M)
RTNH (NeurIPS'22)	4DR	AP_{3D}	24.3	17.2	31.1	59.8	23.7	25.2	33.2	38.6	17.35
		AP_{BEV}	43.0	35.1	56.2	87.1	40.5	42.8	63.2	51.0	
RTNH-DA3D		AP_{3D}	**29.2**	**19.3**	**41.7**	**71.6**	**26.1**	**34.6**	**38.9**	**39.6**	21.97
		AP_{BEV}	**47.0**	**38.5**	**57.3**	**91.1**	**43.1**	**47.4**	**64.4**	**53.4**	
3D-LRF (CVPR'24)	L+4DR	AP_{3D}	50.7	51.6	48.6	74.4	51.1	37.9	61.9	41.7	34.87
		AP_{BEV}	73.3	**75.4**	85.7	89.3	72.1	50.0	85.2	52.3	
3D-LRF-DA3D		AP_{3D}	**54.5**	**51.7**	**60.4**	**81.0**	**56.8**	**42.0**	**64.3**	**43.4**	43.71
		AP_{BEV}	**76.5**	75.3	**89.5**	**92.2**	**77.8**	**51.3**	**87.2**	**53.7**	
L4DR (AAAI'25)	L+4DR	AP_{3D}	53.8	51.6	64.7	78.7	54.6	**48.2**	53.0	40.3	62.04
		AP_{BEV}	77.5	76.5	88.7	90.0	78.4	57.8	88.6	52.8	
L4DR-DA3D		AP_{3D}	**61.9**	**58.9**	**66.4**	**79.2**	**63.0**	**48.2**	**64.6**	**47.6**	88.74
		AP_{BEV}	**78.5**	**77.4**	**89.1**	**90.1**	**79.3**	**58.8**	**88.9**	**60.6**	

Table 2: Comparison of AP_{3D} and AP_{BEV} at IoU threshold 0.3 on the K-Radar benchmark. Results are reported for baseline models and DA3D on RTNH, 3D-LRF, and L4DR backbones under seven different weather conditions.

Method	Modality	Metric	Total	Normal	Overcast	Fog	Rain	Sleet	Li.Snow	He.Snow	Param(M)
RTNH (NeurIPS'22)	4DR	AP_{3D}	48.9	42.8	57.2	83.8	42.1	45.8	62.6	54.5	17.35
		AP_{BEV}	55.1	48.8	65.4	90.6	50.2	52.3	70.5	57.3	
RTNH-DA3D		AP_{3D}	**53.0**	**48.1**	**60.5**	**88.3**	**46.4**	**55.2**	**64.6**	**56.6**	21.97
		AP_{BEV}	**57.2**	**52.2**	**66.9**	**91.6**	**52.9**	**56.5**	70.0	**59.1**	
3D-LRF (CVPR'24)	L+4DR	AP_{3D}	77.8	82.3	86.3	89.8	76.8	53.1	**88.2**	54.0	34.87
		AP_{BEV}	81.1	85.7	91.0	90.2	80.5	57.9	**88.6**	**58.6**	
3D-LRF-DA3D		AP_{3D}	**81.0**	**82.4**	**89.7**	**92.9**	**81.4**	**55.2**	**88.2**	**55.4**	43.71
		AP_{BEV}	**84.3**	**88.1**	**95.9**	**93.0**	**85.3**	**62.6**	**88.6**	58.3	
L4DR (AAAI'25)	L+4DR	AP_{3D}	77.9	77.6	87.9	**89.2**	78.4	59.8	80.3	51.5	62.04
		AP_{BEV}	79.5	85.3	**90.1**	90.0	80.7	**66.6**	89.1	59.6	
L4DR-DA3D		AP_{3D}	**79.3**	**85.9**	**88.4**	**89.2**	**79.7**	**65.8**	**89.0**	**60.2**	88.74
		AP_{BEV}	**80.4**	**86.5**	89.8	**90.1**	**81.0**	62.6	**89.9**	**61.9**	

across 58 driving sequences. Its comprehensive coverage of seven typical weather conditions (normal, rain, sleet, fog, overcast, light snow, and heavy snow) provides an ideal testbed for evaluating domain adaptation and robustness under adverse environments. Following the official split, we use 17,458 frames for training and 17,536 frames for testing. We adopt standard 3D object detection metrics, including AP_{3D} and AP_{BEV} of the "Sedan" class at IoU=0.5 and IoU=0.3, and report results on the official test set across all weather conditions.

To validate the generality of DA3D, we integrate it into three representative 3D detection backbones, covering both radar-only and LiDAR-Radar fusion paradigms. These include the radar-only detector RTNH [25], the LiDAR-Radar fusion model 3D-LRF [7] with spatial feature fusion and weather-aware radar gating, and the advanced multimodal fusion model L4DR [14] with multimodal encoding and foreground-aware denoising, representing leading designs for robust 3D detection under diverse conditions.

Implementation Details. For all baselines, we adopt the official implementations[1] and training protocols. DA3D is inserted in a plug-and-play manner into all convolutional layers without modifying the original network architecture. We employ a two-stage training strategy: first training the backbone on mixed-weather data, followed by freezing the backbone and fine-tuning domain-specific LoRA modules. Each weather domain maintains an independent set of LoRA parameters (rank=32). To ensure fairness, we re-train all baseline models with extended training epochs to match the total training time of DA3D, avoiding performance gaps caused by insufficient training and ensuring that the improvements come from our method design rather than a longer training time.

4.2 Main Results

Table 1 and Table 2 report 3D detection performance of DA3D compared to three baselines—RTNH [25], 3D-LRF [7], and L4DR [14].

[1]https://github.com/kaist-avelab/K-Radar

Under the relaxed IoU=0.3 threshold, DA3D achieves AP_{3D} gains of +4.1%, +3.2%, and +1.4% on RTNH, 3D-LRF, and L4DR, respectively. The improvement becomes more pronounced at the stricter IoU=0.5 threshold, with corresponding gains of +4.9%, +3.8%, and +8.1%. Similar trends are observed in AP_{BEV}. The larger gains at higher IoU reflect DA3D's ability to improve precise object localization, which is more sensitive to domain shift. These results confirm that DA3D consistently improves detection accuracy across modalities and backbones.

We further examine performance under different weather conditions to assess the robustness of DA3D. Substantial improvements are observed in adverse scenarios such as Fog (+11.8% AP_{3D} on RTNH) and Rain (+8.4% on L4DR), where sensor signals are heavily degraded. In Sleet, DA3D achieves notable AP_{3D} gains (e.g., +6.0% on L4DR at IoU=0.3), but a slight drop in AP_{BEV} is observed. We attribute this discrepancy to metric sensitivity: AP_{3D} captures improvements in full 3D localization, including vertical accuracy, while AP_{BEV} is more affected by ground-plane distortions caused by sleet. In addition to accuracy gains, DA3D remains highly parameter-efficient. With domain-aware dynamic rank allocation, it introduces only ∼30% additional parameters on average—specifically, +26.7% on RTNH, +25.4% on 3D-LRF, and +43.0% on L4DR, as summarized in Table 1. This lightweight overhead enables scalable domain-aware modeling without duplicating full models for each weather condition, making DA3D well-suited for deployment in real-world, resource-constrained settings.

4.3 Ablation Study

We conduct comprehensive ablation studies to validate the core components of DA3D. Specifically, we investigate: (1) whether explicitly modeling weather domains enhances multi-weather robustness; (2) whether the performance gains stem from our domain-aware design rather than simply increasing model size; and (3) whether adaptive LoRA rank allocation improves performance under a fixed parameter budget.

Comparison of Multi-Weather Modeling Strategies. We compare four representative strategies for handling diverse weather conditions: (1) the **Independent Model (IM)**, which trains a separate model for each domain and serves as an upper bound; (2) **MTHEAD**, which attaches task-specific detection heads to a shared backbone; (3) **MTLoRA**, which introduces weather-specific LoRA branches with fixed rank; and (4) our proposed **DA3D**, which further incorporates adaptive rank allocation for domain-aware parameterization. We summarize the empirical observations from Tables 3–5. The Independent Model (IM) consistently achieves the highest accuracy across all backbones, as it fully decouples the domains and learns separate parameters for each weather condition. However, this performance comes at the cost of substantially increased model size, reaching over 5× the parameters of DA3D. MTHEAD, which models weather variation only in the detection head, provides limited improvements over the backbone, indicating that shallow domain-specific modeling is insufficient to handle severe domain shifts. MTLoRA alleviates this issue by introducing domain-specific LoRA branches, achieving noticeable gains over MTHEAD, especially at higher IoU thresholds. Finally, DA3D consistently outperforms MTHEAD and MTLoRA across all backbones.

Table 3: Comparison of four different multi-weather modeling strategies on RTNH.

Method	IoU 0.3		IoU 0.5		Param(M)
	AP_{3D}	AP_{BEV}	AP_{3D}	AP_{BEV}	
RTNH	48.9	55.1	24.3	43.0	17.3
IM	**53.3**	**60.0**	<u>29.1</u>	**48.2**	121.4
MTHEAD	49.8	56.0	24.9	43.6	17.6
MTLoRA	51.0	56.7	28.3	45.5	21.9
DA3D (ours)	<u>53.0</u>	<u>57.2</u>	**29.2**	<u>47.0</u>	21.9

Table 4: Comparison of four different multi-weather modeling strategies on 3D-LRF.

Method	IoU 0.3		IoU 0.5		Param(M)
	AP_{3D}	AP_{BEV}	AP_{3D}	AP_{BEV}	
3D-LRF	77.8	81.1	50.7	73.3	34.8
IM	**81.1**	**84.5**	**55.9**	**76.6**	244.1
MTHEAD	78.0	82.3	50.6	74.2	35.1
MTLoRA	80.5	83.3	53.8	75.9	43.7
DA3D (ours)	<u>81.0</u>	<u>84.3</u>	<u>54.5</u>	<u>76.5</u>	43.7

Table 5: Comparison of four different multi-weather modeling strategies on L4DR.

Method	IoU 0.3		IoU 0.5		Param(M)
	AP_{3D}	AP_{BEV}	AP_{3D}	AP_{BEV}	
L4DR	77.9	79.5	53.8	77.5	62.0
IM	<u>79.0</u>	**80.7**	**61.9**	**78.6**	434.0
MTHEAD	78.2	79.6	54.3	77.6	62.8
MTLoRA	78.8	<u>80.4</u>	57.3	78.1	88.7
DA3D (ours)	**79.3**	<u>80.4</u>	**61.9**	<u>78.5</u>	88.7

On the lightweight RTNH, DA3D matches or slightly surpasses IM with only a fraction of the parameters, though the performance gap is marginal, suggesting that the backbone capacity becomes the dominant bottleneck. On larger backbones like 3D-LRF and L4DR, DA3D approaches or even matches IM performance with significantly fewer parameters, demonstrating the increasing advantage of adaptive capacity allocation as the model size increases.

Fair Comparison with Consistent Parameters. To further verify that the performance gain of DA3D does not simply result from increasing model size, we compare DA3D with scaled-up variants of each backbone (`-Large`) under similar parameter budgets. Specifically, the `-Large` models are constructed by uniformly enlarging the channel width of all convolutional layers while keeping the overall network architecture unchanged, serving as a strong baseline to evaluate whether naive parameter scaling alone can close the performance gap. As shown in Table 6, DA3D consistently outperforms both the original and `-Large` backbones across all settings, despite having comparable or even fewer parameters. The performance gain is particularly evident on the lightweight RTNH, where increasing the model size brings only marginal improvements, whereas DA3D achieves substantial gains with fewer parameters. On larger backbones like 3D-LRF and L4DR, the advantage of DA3D becomes even more pronounced, demonstrating that our adaptive rank allocation and domain-aware design better utilize the available capacity compared to naive scaling.

Table 6: Comparison with parameter-scaled backbones under similar parameter budgets. The -Large models are obtained by uniformly increasing the channel width of all convolutional layers.

Method	IoU 0.3		IoU 0.5		Param(M)
	AP_{3D}	AP_{BEV}	AP_{3D}	AP_{BEV}	
RTNH	48.2	56.7	24.3	43.0	17.3
RTNH-Large	50.6	56.6	26.6	46.0	22.6
RTNH-DA3D	**53.0**	**59.9**	**29.2**	**47.0**	21.9
3D-LRF	77.7	81.1	50.7	73.3	34.8
3D-LRF-Large	78.1	81.2	52.5	73.9	44.1
3D-LRF-DA3D	**81.0**	**84.3**	**54.5**	**76.5**	43.7
L4DR	78.0	79.5	53.5	77.5	62.8
L4DR-Large	78.4	80.0	56.8	77.3	88.8
L4DR-DA3D	**79.3**	**80.4**	**61.9**	**78.5**	88.7

Effectiveness of Adaptive Rank Allocation. We further evaluate the effectiveness of adaptive rank allocation by comparing DA3D with a fixed-rank variant (MTLoRA) under the same total LoRA capacity. Specifically, we control the per-domain per-layer rank r and vary it from 8 to 48. In DA3D, although the rank $r_w^{(l)}$ for each domain is dynamically adjusted, we constrain the total rank within each layer, $R_{\text{total}}^{(l)}$, to remain constant. This enables DA3D to flexibly allocate capacity across domains without increasing parameters. As shown in Fig. 3, DA3D consistently outperforms MTLoRA in both AP_{3D} and AP_{BEV}, especially under low-to-medium ranks ($r = 8$–32), where adaptive allocation brings significant gains. As r increases, the performance gap narrows as both methods have sufficient capacity. These results demonstrate the advantage of adaptive rank allocation for robust 3D detection in resource-constrained settings.

5 Conclusion

We present DA3D, a domain-aware and lightweight adaptation framework that explicitly addresses the domain shift problem in all-weather 3D object detection. While existing methods often assume that 4D radar is inherently robust to adverse weather, they largely overlook the domain shift induced by different weather conditions in both radar-only and LiDAR–Radar fusion settings. To tackle this challenge, DA3D inserts domain-specific LoRA modules into both the 3D sparse encoder and 2D BEV encoder, enabling weather-aware feature modulation within a unified backbone at minimal parameter cost. Furthermore, we propose a dynamic rank adaptation strategy that reallocates LoRA capacity based on domain difficulty and gradient sensitivity, allowing efficient parameter utilization under a fixed budget. Extensive experiments on the K-Radar benchmark validate the superiority of DA3D over state-of-the-art multi-weather modeling methods, including multi-head and fixed-rank LoRA baselines. DA3D consistently improves 3D detection accuracy across both radar-only and LiDAR–Radar fusion detectors, with notable gains in adverse conditions such as fog and snow.

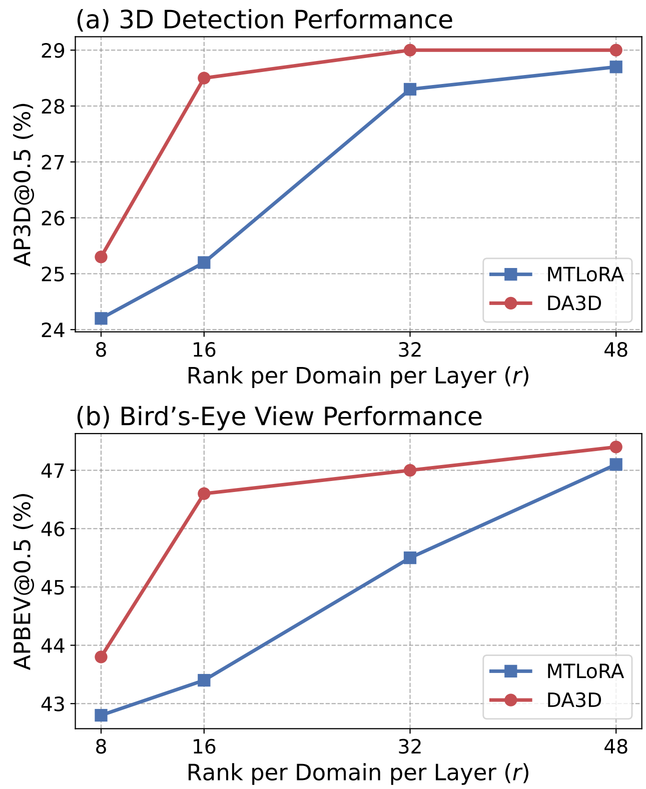

Figure 3: Performance comparison of DA3D and MTLoRA on RTNH under varying per-domain per-layer LoRA ranks (r) from 8 to 48. Both methods use a fixed total LoRA rank per layer $R_{\text{total}}^{(l)}$, where DA3D dynamically allocates ranks across domains, while MTLoRA assigns equal ranks to all domains.

Limitations. While DA3D effectively mitigates weather-induced domain shift with minimal overhead, it still faces several challenges. The storage cost of domain-specific LoRA modules grows linearly with the number of weather domains, and DA3D relies on a predefined set of discrete weather labels and a camera-based weather classifier for domain routing. This may limit its applicability under continuous domain shifts or sensor-constrained settings. Future work includes exploring domain-agnostic adaptation modules and continuous domain generalization beyond predefined weather categories to further enhance scalability and robustness under diverse real-world conditions.

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